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MASS AUGMENTATION OF A TENSION-LEG PLATFORM

Abstract

Tension-leg platforms for supporting wind turbines are augmented with surge plates. The surge plates increase the amount of water that is displaced when a tension-leg platform is accelerated horizontally, which reduces wave induced accelerations. The surge plates are mounted to the deepest parts of the submerged structure of each platform to minimize wave loading.

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Background/Summary

BACKGROUND

[0001] As the offshore wind industry moves into deeper water, designers have been searching for

the most economical solution to support wind turbines on floating platforms. One solution, Tension-Leg Platforms (TLPs) shows promise in reducing the cost of floating offshore wind. TLPs have been used in the offshore Oil and Gas (O&G) industry since the 1980s and have particular traits which make them well suited to support wind turbines. However, TLPs have only recently started to receive the attention needed to optimize them for supporting wind turbines.

[0002] A TLP is an offshore structure comprising a hull with net buoyancy, secured to the seabed by pretensioned tendons. The tendons are secured to the seabed or the floor of any other body of water with anchors. The tendons attach to the platform at the tendon porch. The tension in the tendons counteracts the hull's net buoyancy, which stabilizes the hull. The required pretension in the tendons is determined by the design loads on the platform; primarily environmental loads from wind and waves. The larger the design loads, the higher the pretension must be to maintain the platform's stability.

[0003] For a certain combination of wind turbine and site conditions the wind loads on a TLP designed to support a wind turbine are nominally fixed. This means that if wave loads are ignored, the platform has baseline requirements for: [0004] minimum tendon pretension: to withstand the overturning moments from the wind, [0005] minimum structural capacity: to withstand the wind loads, [0006] minimum buoyant volume: to provide the net buoyancy to float the mass of the platform, tower, and wind turbine, and achieve the minimum tendon pretension.

[0007] Loads from waves will increase these baseline limits because the wave loads will interact with the buoyant volume. The wave loads on the submerged body of the platform will increase platform motions which will reduce the stability and require a higher tendon pretension. Additionally, the increased motions will increase internal structural loads which will require the structure to be strengthened. A higher pretension requirement and higher structural mass will increase the required buoyant volume, which will attract more wave loads. If a designer does not take care to properly optimize a TLP to support a wind turbine it is easy to see how the introduction of wave loads could cause the design to spiral out of control.

[0008] TLPs for the O&G industry are designed to support massive topside drilling rigs, which results in platform designs that are very large and heavy. However, TLPs designed to support wind turbines can be over 10 times smaller and lighter due to the small tower footprint and comparatively light tower and turbine mass. This means that TLPs designed to support wind turbines will have a different dynamic response compared to O&G TLPs. However, while the dynamics are different, existing designs for TLPs supporting wind turbines generally follow the same design principles used on O&G TLPs. There is room to improve the designs of TLPs supporting wind turbines.

[0009] Per Newton's second law of motion, a floating structure will accelerate as a function of the magnitude of the applied wave force and the mass of the structure. With the accelerations of the structure being directly proportional to the internal structural loads, a primary goal of TLP designs is to minimize the wave induced accelerations. Minimizing the accelerations will result in lower internal loads enabling the design of lower cost platforms.

[0010] TLPs are typically designed to minimize accelerations by minimizing wave forces; attempting to be transparent to the waves. This is done by minimizing the water plane area and buoyant volume near the water surface. Wave particle velocities decay exponentially with water depth. The highest wave forces will therefore be generated near the surface of the water and decrease quickly with depth. By minimizing the amount of structure near the surface and moving the bulk of the structure deeper under the water, the total applied wave forces can be minimized.

[0011] TLP designs could also reduce accelerations by increasing platform mass. However, to maintain the required pretension the buoyant volume would need to be increased which would attract more wave loads and cancel out the reduction in accelerations from the increased mass. Consequently, TLPs designed to support wind turbines have focused solely on reducing wave loads and neglected to consider the potential benefits of augmenting the mass of the platforms. The

purpose of this disclosure is to describe a method of augmenting the mass of TLPs designed to support wind turbines which helps to reduce the accelerations on the platforms and therefore lower the cost of existing platform designs.

SUMMARY

[0012] The present disclosure can improve upon the basic design of a Tension-Leg platform by augmenting the mass of the platform in the horizontal direction. The present disclosure involves attaching surge plates to the submerged structure of a TLP to displace water when the platform accelerates horizontally, thereby increasing the horizontal added mass. It is envisioned that the surge plates will be thin flat plates oriented substantially vertical when the platform is upright and optimally located deep under water to increase the horizontal added mass of the platform while attracting minimal wave loading. The size and placement of the surge plates can be tuned to optimally augment the surge added mass.

[0013] Whereas it is typical to use the terms surge and sway to denote the fore-aft and side-side motions respectively of a ship, the directional terms are simplified here such that surge denotes any direction parallel to the still water.

[0014] In the field of fluid mechanics, added mass is defined as the mass of water surrounding a submerged body which is displaced (accelerated) when the body is accelerated relative to the surrounding water. When a submerged structure accelerates, it will displace the surrounding water such that the water moves out of the way in front of the structure, and fills in behind the structure. To accelerate a platform in surge, a unit force must act upon the effective horizontal mass of the platform, which is the sum of the mass of the platform plus the added mass of the displaced water. Consequently, by displacing more water, surge plates increase the effective horizontal mass of the platform which reduces horizontal accelerations, thereby reducing internal loads on the structure. By reducing internal loads in the platform, the structure can be built lighter and cheaper, resulting in a more cost-effective design.

[0015] The present disclosure benefits from the fact that added mass does not affect the pretension of the tendons. Tendon pretension is calculated in the static condition when neither the platform nor the surrounding water is accelerating. Without a relative acceleration between the platform and surrounding water the added mass is zero in all directions. Consequently, the added mass does not augment the static mass of the platform and pretension remains unaffected. Adding fixed mass to the platform, be it structural or ballast, would increase the weight of the platform and decrease the tendon pretension. This would require the buoyant volume of the platform to be increased to counteract the additional mass which would increase wave loads on the primary structure. However, since the added mass doesn't affect the static pretension, the buoyant volume remains unchanged and the wave loads on the primary structure remain unchanged. Therefore, surge plates provide a novel solution to reduce platform accelerations by increasing a TLP's effective horizontal mass without affecting buoyant volume or pretension.

[0016] It is unavoidable that the surge plates will attract additional wave loads that could counteract the reduction in accelerations provided by the increase in added mass. However, it is possible to minimize the additional wave loads by taking advantage of the exponential decay of wave particle velocities with respect to water depth. Just as existing TLPs are designed to minimize the amount of structure near the free surface by locating the bulk of the structure deeper underwater to minimize wave loading, this same design principle can be applied to surge plates. By attaching surge plates near, or even below, the base of the submerged structure the wave loads on the surge plates can be minimized. Consequently, when surge plates are properly positioned on a TLP deep under water, the increase in effective mass well exceeds the increase in wave loads, resulting in a net reduction in platform accelerations.

[0017] There are two main reasons that surge plates are not beneficial for an O&G TLP. First, O&G TLPs are already very heavy, meaning that there isn't a significant benefit to increasing the effective mass through added mass augmentation. Second, O&G platforms are concerned with

limiting surge offset so that riser angles can be minimized. Risers are connected vertically between a platform down to the seabed. As a platform surges away from its static position it forces the riser to bend and take on an “S” shape. The bending introduces stresses which can cause the riser to fail. Horizontal loads on an O&G platform from ocean currents would be increased if surge plates were added, increasing the maximum surge offset and riser angle, which is not desirable. Surge plates provide a novel means to augment the added mass on TLPs designed to support a wind turbine, because they are significantly lighter than O&G TLPs and do not have a surge offset limitation. [0018] The present disclosure includes a method of decreasing wave induced accelerations of a tension-leg platform by installing additional secondary structures (plates and/or panels) to form the surge plates.

[0019] The method includes increasing a mass of water displaced by the platform when the platform is accelerating horizontally by a system of one or more surge plates connected to a submerged structure of the platform, the surge plates including thin plates and/or panels, oriented to be substantially vertical when the platform is upright, whereby a submerged lateral projected area of the surge plates is greater than or equal to 10% of a lateral projected area of the submerged structure of the platform excluding the surge plates.

[0020] There is no requirement that the surge plates be constructed from any specific material(s), but could include materials such as metals, plastics, concrete, or a combination thereof. While these structures add to the mass and cost of the platform, these increases are offset by reductions in the mass and cost of the primary structure. By reducing the accelerations and resulting internal loads on the primary structure surge plates enable the design of TLPs which have lower mass and cost than the base designs without surge plates.

[0021] This summary is provided to introduce a concept in a simplified form that is further described below in the Detailed Description. This summary is not intended to identify key features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

DESCRIPTION OF THE DRAWINGS

[0022] The foregoing aspects and many of the attendant advantages of this disclosure will become more readily appreciated as the same become better understood by reference to the following detailed description, when taken in conjunction with the accompanying drawings, wherein:

[0023] FIG. 1 is a diagrammatical illustration of a tension-leg platform with surge plates, the illustration depicts the exponential decay of wave particle orbital velocities surrounding a TLP and the respective lateral projected areas of the platform and surge plates;

[0024] FIG. 2 is a diagrammatical illustration of two sections taken through the arms of TLPs depicted in FIGS. 3 and 4 showing surge plates attached to the structure of the arms and the increase in added mass of the arms due to the surge plates;

[0025] FIG. 3 is a diagrammatical illustration of a tension-leg platform with central column, five tubular truss arms, and surge plates mounted internal to the structural envelope within the arm trusses;

[0026] FIG. 4 is a diagrammatical illustration of a tension-leg platform with five down-sloping box arms, and a surge plate mounted external to the structural envelop on top of each arm;

[0027] FIG. 5 is a diagrammatical illustration of a tension-leg platform with a central column, five up-sloping box arms, and a surge plate mounted external to the structural envelope below each arm;

[0028] FIG. 6 is a diagrammatical illustration of a pyramidal shaped tension-leg platform with surge plates mounted external to the structural envelope to the bottom of the platform;

[0029] FIG. 7 is a diagrammatical illustration of a tension-leg platform with central column, 3

cylindrical arms, and surge plates mounted internal to the structural envelope, below the diagonal bracing;

[0030] FIG. **8** is a diagrammatical illustration of a tension-leg platform with a square base and surge plates mounted both internal and external to the structural envelope, above and below the base structure of the platform;

[0031] FIG. **9A** is a diagrammatical illustration of a constant diameter central column;

[0032] FIG. **9B** is a diagrammatical illustration of a variable diameter central column;

[0033] FIG. **10** is a diagrammatical illustration of a vertically oriented planar tubular truss arm with attached surge plates;

[0034] FIG. **11** is a diagrammatical illustration of a down-sloping box arm with tapered sides and a surge plate attached to the top of the arm;

[0035] FIG. **12** is a diagrammatical illustration of a down-sloping box arm with parallel sides and a surge plate attached to the top of the arm;

[0036] FIG. **13** is a diagrammatical illustration of an up-sloping box arm with tapered sides and a surge plate attached to the bottom of the arm with a keel plate attached along the bottom edge of the surge plate;

[0037] FIG. **14** is a diagrammatical illustration of an up-sloping box arm with parallel sides and a surge plate attached to the bottom of the arm with a keel plate attached along the bottom edge of the surge plate;

[0038] FIG. **15** is a diagrammatical illustration of an example of a porous surge plate;

[0039] FIG. **16A** is a diagrammatical illustration of an example of a surge plate with no enclosed buoyant volume;

[0040] FIG. **16B** is a diagrammatical cross-sectional illustration of the surge plate of FIG. **16A**;

[0041] FIG. **17A** is a diagrammatical illustration of an example of a surge plate with an enclosed buoyant volume; and

[0042] FIG. **17B** is a diagrammatical cross-sectional illustration of the surge plate of FIG. **17A**.

DETAILED DESCRIPTION

[0043] When a submerged structure accelerates in a fluid, the surrounding fluid will be displaced such that it moves out of the way in front of the structure and fills in behind the structure. Because the fluid must be accelerated to be displaced, the total mass of the displaced fluid adds to the effective inertia of the submerged structure. In the field of fluid mechanics, the mass of the displaced fluid is known as added mass.

[0044] In one embodiment, surge plates are attached to a Tension-Leg Platform (TLP) designed to support offshore wind turbines. In one embodiment, the surge plates consist of thin plates and/or panels, oriented to be substantially vertical when the platform is upright, whereby the submerged lateral projected area of the surge plates is greater than or equal to 10% of the lateral projected area of the submerged structure of the platform excluding the surge plates. The surge plates may be configured to be open structures with no enclosed buoyant volumes or closed structures with enclosed buoyant volumes. FIGS. **16A** and **16B** illustrate an example of a surge plate **1602** with no enclosed buoyant volume, i.e., a solid interior **1604**. FIGS. **17A** and **17B** illustrate an example of a surge plate **1702** with an enclosed buoyant volume **1704**. The surge plates may also be configured to be nonporous allowing no water to pass through the surge plates or porous allowing some water to pass through openings in the surge plates. FIG. **15** is an example of a porous surge plate **1502** with holes **1504**. Holes **1504** in the surge plates do not contribute to the lateral projected area of the surge plates. FIG. **1** depicts the above lateral projected areas of one embodiment of a TLP **100** with surge plates **102**. Reference **104** of FIG. **1** depicts the exponential decay of wave particle orbital velocities surrounding a TLP and the respective lateral projected areas **108**, **106** of the platform **100** and surge plates **102**. “Lateral projected area” has the common ordinary meaning of being a two dimensional area of a three dimensional structure, when the three dimensional structure is projected onto any vertical plane. The submerged lateral projected area of the platform **100** visible in FIG. **1**

including the arms **106**, and the part of the column **108** that is under the surface of the water **126**. is denoted by the single cross-hatching. The lateral projected area of the surge plates **102** visible in FIG. **1** is denoted by the double cross-hatching. Whereas it is typical to use the terms surge and sway to denote the fore-aft and side-side motions respectively of a ship, the directional terms are simplified here such that surge denotes any direction parallel to the still water (horizontal).

[0045] By increasing the lateral projected area of the submerged structure, the surge plates increase the amount of water that is displaced when the TLP is accelerating horizontally relative to the water. This increases the horizontal added mass of the TLP, thereby increasing the effective horizontal mass of the platform.

[0046] Whereas the displaced water is flowing freely around the submerged structure it is typical to represent the added mass as a fixed volume that moves with the structure.

[0047] FIG. **2** shows a diagrammatical illustration of two section cuts through the arms of TLPs **300** and **400** illustrated in FIGS. **3** and **4**. The areas filled with dashed lines in the diagram denote the primary structure of the arms **306**, **308**, and **404**. The single hatching **202** denotes cross-sections of the horizontal added mass of water surrounding the primary arm structures that would be displaced by the arms if surge plates were not installed. The double hatching **204** denotes the additional added mass of water that would be displaced if surge plates **302**, and **402** were installed.

[0048] In way of the section illustrated in FIG. **2** for the tubular truss arm **314** of FIG. **3**, the surge plate **302** attached between the top and bottom chords **306**, **308** increases the cumulative vertical length of the section by 72%, which approximately increases the total added mass of the section by 340%. In way of the section illustrated in FIG. **2** for the box arm **404**, the surge plate **402** attached to the top of the arm increases the cumulative vertical length of the section by 68%, which approximately increases the total added mass of the section by 130%.

[0049] In one embodiment, the surge plates are attached to the lower half of the submerged structure of the TLP. TLPs are moored to the seabed with fixed length tendons meaning their draft varies with the tides. For the present disclosure, the submerged structure refers to the structure submerged at low astronomical tide. "Low Astronomical Tide" has the common ordinary meaning of the lowest predicted still water level due to combined tidal action of the sun and moon. As depicted in FIG. **1**, wave particle orbital velocities **104** decrease exponentially with respect to water depth. By attaching the surge plates to the deepest parts of the submerged structure the particle velocities will be minimized; thereby minimizing the wave loads on the surge plates.

[0050] In one embodiment, the TLP to which the surge plates are attached comprises a central column with five arms extending radially from the base of the central column. It is to be understood that the particular TLP in the figures is for illustrating aspects of the disclosure which may be applicable to other types of TLPs. For example, FIGS. **3**, **4**, **5**, **6**, **7**, and **8** illustrate TLPs which may be utilized in the present disclosure. Therefore, it is to be further understood that the disclosure can apply to any platform with positive net buoyancy that uses pretensioned tendons for anchoring the platform in or on a body of water.

[0051] FIG. **3** shows a diagrammatical illustration of one embodiment of a TLP **300** designed to support a wind turbine, with surge plates **302**, **304** installed to increase the horizontal added mass of the platform. The TLP **300** takes the form of a central cylindrical column with five arms **314**, equally spaced circumferentially around and extending radially from the base of the central column. The column is constructed of two cylinders **316**, **318** of different diameters serially connected by a tapered conical section **320**, the bottom most cylinder **316** having the larger diameter and the diameter of the upper cylinder **318** having the smaller diameter. Each arm **314** is arranged with a vertically oriented cylindrical buoyancy can **312** supported at a distance radially from the central column **316** by a vertically oriented planar truss. The top chord **306** and the bottom chord **308** of the truss are constructed from large-diameter cylindrical pipes, with each chord having a first end for connecting to the central column **316** and a second end for connecting to the buoyancy can **312**. The chords **306**, **308** are supported near their mid-spans by a single cylindrical

vertical cylinder **312**. At the distal end of each arm one or more tendon porches **322** are attached. Once the platform is installed, the base of the central column and arms are located a distance below the water surface **326**. The central column extends a distance above the water surface **326** and supports the base of the wind turbine tower **324**.

[0052] In one embodiment, two surge plates **302**, **304** are attached to each arm **314** internal to the structural envelope, filling the open spaces inside each truss. The four sides of the inner surge plate **302** are attached to the side of the central column **316**, the underside of the top chord **306**, the top of the bottom chord **308** and the inside edge of the vertical cylinder **310** for each respective arm **314**. The four sides of the outer surge plate **304** are attached to the underside of the top chord **306**, the top of the bottom chord **308**, the outside edge of the vertical cylinder **310** and the inside edge of the buoyancy can **310** for each respective arm. In one embodiment, the total lateral projected area of the surge plates **302**, **304** is equal to 15.7% of the submerged lateral projected area of the platform (e.g., **306**, **308**, **310**, **312**, **316**, **318**, **320**, **322**) without surge plates. In alternate embodiments the total lateral projected area of the surge plates **302**, **304** may be greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **306**, **308**, **310**, **312**, **316**, **318**, **320**, **322**) without surge plates. In one embodiment, the total lateral projected area of the two surge plates **302**, **304** is equal to 53% of the lateral projected area of the arm **314** without the two surge plates. In alternate embodiments the total lateral projected area of one or more surge plates is greater than or equal to 25% of the lateral projected area of the arm **314** without the surge plate(s). In one embodiment, the lateral projected area of each surge plate **302**, **304** is greater than or equal to 50% of the lateral projected area of the respective open area within the arm truss.

[0053] FIG. 4 shows a diagrammatical illustration of one embodiment of a TLP **400** designed to support a wind turbine, with surge plates **402** installed to increase the horizontal added mass of the platform. The TLP takes the form of a central cylindrical column with five arms **404**, equally spaced circumferentially around and extending radially from the base of the central column. The column is constructed of two cylinders **416**, **418** of different diameters serially connected by a tapered conical section **420**, the bottom most cylinder **416** having the larger diameter and the diameter of the upper cylinder **418** having the smaller diameter. Each arm **404** includes a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm **404** are constructed of flat panels. The bottom of the arm **404** is aligned with the bottom of the platform and parallel to baseline. The top of the arm **404** slopes down such that the tip of the arm **404** is shorter in height than the root of the arm **404**. The sides of each arm **404** are angled inwards such that the tip of each arm **404** is narrower than its root. Once the platform is installed, the base of the central column and arms are located a distance below the water surface **430**. The central column extends a distance above the water surface **430** and supports the base of the wind turbine tower **424**.

[0054] In one embodiment, a single surge plate **402** is attached externally to the structural envelope, on top of each arm **404**. Each surge plate **402** forms a triangle, with the hypotenuse attached to the top of the arm **404**, the long edge **426** running parallel to baseline and the short edge **428** extending vertically upwards from the tendon porch **422**. In one embodiment, the total lateral projected area of the surge plates **402** is equal to 17.3% of the submerged lateral projected area of the platform (e.g., **404**, **416**, **418**, **420**, **422**) without surge plates. In alternate embodiments, the total lateral projected area of the surge plates **402** is greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **404**, **416**, **418**, **420**, **422**) without surge plates. In one embodiment, the lateral projected area of the surge plate **402** is equal to 60% of the lateral projected area of the arm **404** without surge plates. In alternate embodiments, the lateral projected area of the surge plate **402** is greater than or equal to 25% of the lateral projected area of the arm **404** without surge plates.

[0055] FIG. 5 shows a diagrammatical illustration of one embodiment of a TLP **500** designed to support a wind turbine, with surge plates **502** installed to increase the horizontal added mass of the

platform. The TLP takes the form of a central cylindrical column with five arms **504**, equally spaced circumferentially around and extending radially from the base of the central column. The column is constructed of two cylinders **516**, **518** of different diameters serially connected by a tapered conical section **520**, the bottom most cylinder **516** having the larger diameter and the diameter of the upper cylinder **518** having the smaller diameter. Each arm **504** includes a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm **504** are constructed of flat panels. The top of the arm **504** is aligned parallel to the baseline of the platform at a distance above baseline. The bottom of the arm **504** slopes up from baseline such that the tip of the arm **504** is shorter in height than the root of the arm **504**. The sides of each arm **504** are angled inwards such that the tip of each arm **504** is narrower than its root. Once the platform is installed, the base of the central column and arms are located a distance below the water surface **530**. The central column extends a distance above the water surface **530** and supports the base of the wind turbine tower **524**.

[0056] In one embodiment, a single surge plate **502** is attached externally to the structural envelope, to the bottom of each arm **504**. Each surge plate **502** forms a triangle, with the hypotenuse attached to the bottom of the arm **504**, the long edge **526** running parallel to baseline and the short edge **528** extending vertically downwards from the tendon porch **522**. A keel plate **532** is attached to each surge plate **502** along the long edge **526** of the triangle. In one embodiment, the total lateral projected area of the surge plates **502** is equal to 17.3% of the submerged lateral projected area of the platform (e.g., **504**, **516**, **518**, **520**, **522**) without surge plates. In alternate embodiments, the total lateral projected area of the surge plates **502** is greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **504**, **516**, **518**, **520**, **522**) without surge plates. In one embodiment, the lateral projected area of the surge plate **502** is equal to 60% of the lateral projected area of the arm **504** without surge plates. In alternate embodiments, the lateral projected area of the surge plate **502** is greater than or equal to 25% of the lateral projected area of the arm **504** without surge plates.

[0057] FIG. **6** shows a diagrammatical illustration of one embodiment of a TLP **600** designed to support a wind turbine, with surge plates **602** installed to increase the horizontal added mass of the platform. The TLP takes the form of a hollow tetrahedron; the six edges of the pyramid are formed from cylindrical members **604**, **606**, **608**, **610**, **612**, and **614**. The triangular base of the tetrahedron is formed from cylindrical members **610**, **612**, and **614**. Cylindrical members **604**, **606**, and **608** form the edges of the tetrahedron which connect between the vertices of the base to the apex of the tetrahedron. One or more tendon porches **622** are attached at each vertex of the base of the pyramid. Once the platform is installed, the triangular base of the tetrahedron is located a distance below the surface of the water **626**. The edges **604**, **606**, **608** of the tetrahedron converge at the apex, which is located a distance above the surface of the water **626**. At the apex, the cylindrical members **604**, **606**, **608** converge into the central cylindrical transition piece **620** which supports the base of the wind turbine tower **624**.

[0058] In one embodiment, surge plates **602** are attached to the external structural envelope of the TLP. The surge plates **602** are attached to the underside of the cylinders **610**, **612**, **614** forming the base of the pyramid. The surge plates **602** can extend the full length of the cylinders **610**, **612**, **614**. In one embodiment, the total lateral projected area of the surge plates **602** is equal to 15.7% of the submerged lateral projected area of the platform (e.g., **604**, **606**, **608**, **610**, **612**, **614**) without surge plates. In alternate embodiments, the total lateral projected area of the surge plates **602** is greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **604**, **606**, **608**, **610**, **612**, **614**) without surge plates.

[0059] FIG. **7** shows a diagrammatical illustration of one embodiment of a TLP **700** designed to support a wind turbine, with surge plates **702** installed to increase the horizontal added mass of the platform. The TLP takes the form of a central constant diameter cylindrical column with three arms **714**, equally spaced circumferentially around and extending radially from the base of the central

column. Each of the arms **714** is formed from a single cylinder **706** and supported by a diagonal cylinder **704**. The first end of each diagonal cylinder **704** connecting to each respective arm **714** near their mid span and a second end attached to the central column **716** a distance above the root of the respective arm **714**. At the distal end of each arm **714** one or more tendon porches **722** are attached. Once the platform is installed, the base of the central column **716** and arms **714** are located a distance below the water surface **726**. The central column **716** extends a distance above the water surface **726** and supports the base of the wind turbine tower **724**.

[0060] In one embodiment a surge plate **702** is attached to each arm **714** within the structural envelope of the platform. The three sides of each surge plate **702** are attached to the side of the central column **716**, the top of each respective arm cylinder **706**, and the underside of each respective diagonal cylinder **704**. In one embodiment, the total lateral projected area of the surge plates **702** is equal to 24.6% of the submerged lateral projected area of the platform (e.g., **704**, **706**, **716**, **722**) without surge plates. In alternate embodiments, the total lateral projected area of the surge plates **702** is greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **704**, **706**, **716**, **722**) without surge plates.

[0061] FIG. **8** shows a diagrammatical illustration of one embodiment of a TLP **800** designed to support a wind turbine, with surge plates **802** installed to increase the horizontal added mass of the platform. The TLP takes the form of a hollow prism shaped like a square pyramid stacked on top of a rectangular prism. The twelve edges of the structure are formed from cylindrical members. Four horizontal cylindrical members (e.g., **808**) form the square base of the rectangular prism. The vertical edges of the rectangular prism are formed by four additional cylindrical members (e.g., **806**). The remaining four cylindrical members (e.g., **804**) form the edges of the square pyramid and connect the vertical edges (e.g., **806**) of the rectangular prism to the apex of the square pyramid. One or more tendon porches **822** are attached at each vertex of the base of the structure. Once the platform is installed, the base of the structure is located a distance below the surface of the water **826**. The vertical cylinders **806** extend up through the water surface **826**. At the apex, the cylinders **804** converge into a central cylindrical transition piece **820** which supports the base of the wind turbine tower **824**.

[0062] In one embodiment, surge plates **802**, **812** are attached both internally and externally to the structural envelope of the platform. Surge plates **812** are attached internally to the tops of the cylinders **808** forming the square base of the structure and can extend from one vertical cylinder to another vertical cylinder. Surge plates **802** are attached externally to the underside of the cylinders **808** forming the square base of the structure and can extend the length of the cylinder. In one embodiment, the total lateral projected area of the surge plates is equal to 45.9% of the submerged lateral projected area of the platform (e.g., **806**, **808**) without surge plates. In alternate embodiments, the total lateral projected area of the surge plates is greater than or equal to 10% of the submerged lateral projected area of the platform (e.g., **806**, **808**) without surge plates.

[0063] Multiple embodiments of a TLP comprising a central column with arms extending radially from the base of the central column with surge plates are depicted in FIGS. **3**, **4** and **5**. Additional variations on these embodiments are developed further in FIGS. **9-14**.

[0064] FIGS. **9A** and **9B** depict two embodiments of the central column. FIG. **9A** shows a diagrammatical illustration of one embodiment of a central column for a TLP designed to support a wind turbine, wherein the central column is constructed of a single constant diameter cylinder **902**. FIG. **9B** shows a diagrammatical illustration of one embodiment of a central column for a TLP designed to support a wind turbine, wherein the column is constructed of segmented cylinders **904**, **906** of different diameters. The cylinders **904**, **906** are serially connected by a tapered conical section **908**. The bottom most cylinder **904** has the largest diameter and the diameters of the remaining cylinders **906** incrementally decrease with height.

[0065] FIG. **10** shows a diagrammatical illustration of one embodiment of a tubular truss arm **314** for a TLP designed to support a wind turbine. The arm **314** is arranged with a vertically oriented

cylindrical buoyancy can **312** supported at a distance radially from the central column **316** by a vertically oriented planar truss. The top chord **306** and the bottom chord **308** of the truss are constructed from large-diameter cylindrical pipes, with each chord having a first end for connecting to the central column **316** and a second end for connecting to the buoyancy can **312**. The chords **306** and **308** are supported near their mid-spans by a single cylindrical vertical cylinder **310**. At the distal end of each arm **314** one or more tendon porches **322** are attached. Two surge plates **302**, **304** are attached to the arm **314** internal to the structural envelope, filling the open spaces inside the arm truss **314**. The four sides of the inner surge plate **302** are attached to the side of the central column **316**, the underside of the top chord **306**, the top of the bottom chord **308** and the inside edge of the vertical cylinder **310**. The four sides of the outer surge plate **304** are attached to the underside of the top chord **306**, the top of the bottom chord **308**, the outside edge of the vertical cylinder **310** and the inside edge of the buoyancy can **312**. In one embodiment, the total lateral projected area of the two surge plates **302**, **304** is equal to 53% of the lateral projected area of the arm **314** without the two surge plates. In alternate embodiments, the total lateral projected area of the two surge plates **302**, **304** is greater than or equal to 40% of the lateral projected area of the arm **314** without the two surge plates. In one embodiment, the lateral projected area of each surge plate **302**, **304** is greater than or equal to 90% of the lateral projected area of the respective open area within the arm truss. [0066] FIG. **11** shows a diagrammatical illustration of one embodiment of a box arm **404** for a TLP designed to support a wind turbine. The arm **404** includes a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm **404** are constructed of flat panels. The bottom of the arm **404** is aligned with the bottom of the platform and parallel to baseline. The top of the arm **404** slopes down such that the tip of the arm **404** is shorter in height than the root of the arm **404**. The sides of the arm **404** are angled inwards such that the tip of the arm **404** is narrower than its root. In one embodiment a single surge plate **402** is attached externally to the structural envelope, to the top of the arm **404**. The surge plate **402** forms a triangle, with the hypotenuse attached to the top of the arm **404**, the long side **426** running parallel to baseline and the short side **428** extending vertically upwards from the tendon porch **422**. In one embodiment, the lateral projected area of the surge plate **402** is equal to 60% of the lateral projected area of the arm **404** without surge plates. [0067] FIG. **12** shows a diagrammatical illustration of one embodiment of a box arm **1204** for a TLP designed to support a wind turbine. The arm **1204** includes a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm **1204** are constructed of flat panels. The bottom of the arm **1204** is aligned with the bottom of the platform and parallel to baseline. The top of the arm **1204** slopes down such that the tip of the arm is shorter in height than the root of the arm. The sides of the arm **1204** are parallel such that the tip of the arm is the same width as its root. In one embodiment, a single surge plate **1202** is attached externally to the structural envelope, to the top of the arm **1204**. The surge plate **1202** forms a triangle, with the hypotenuse attached to the top of the arm **1204**, the long side **1226** running parallel to baseline and the short side **1228** extending vertically upwards from the tendon porch **1222**. In one embodiment, the lateral projected area of the surge plate **1202** is equal to 60% of the lateral projected area of the arm **1204** without surge plates. In alternate embodiments, the lateral projected area of the surge plate **1202** is greater than or equal to 25% of the lateral projected area of the arm **1204** without surge plates. [0068] FIG. **13** shows a diagrammatical illustration of one embodiment of a box arm **504** for a TLP designed to support a wind turbine. The arm **504** includes a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm **504** are constructed of flat panels. The top of the arm **504** is parallel to and located a distance above baseline. The bottom of the arm **504** slopes up from baseline such that the tip of the arm **504** is shorter in height than the root of the arm **504**. The sides of the arm **504** are angled inwards such that the tip of the arm **504** is narrower than its root. In one embodiment a single surge plate **502** is attached externally to the structural envelope, to the bottom of the arm **504**. The surge plate **502** forms a triangle, with the hypotenuse attached to the bottom of the arm **504**, the long side **526** running parallel to baseline and the short side **528**

extending vertically downwards from the tendon porch 522. In one embodiment a keel plate 532 is attached to support the long side 526 of the surge plate 502 along baseline. In one embodiment, the lateral projected area of the surge plate 502 is equal to 60% of the lateral projected area of the arm 504 without surge plates. In alternate embodiments, the lateral projected area of the surge plate 502 is greater than or equal to 25% of the lateral projected area of the arm 504 without surge plates. [0069] FIG. 14 shows a diagrammatical illustration of one embodiment of a box arm 1404 for a TLP designed to support a wind turbine. The arm 1404 forms a polyhedron approximately shaped like a truncated wedge. The outer surfaces of the arm 1404 are constructed of flat panels. The top of the arm 1404 is parallel to and located a distance above baseline. The bottom of the arm 1404 slopes up from baseline such that the tip of the arm is shorter in height than the root of the arm. The sides of the arm 1404 are parallel such that the tip of the arm is the same width as its root. In one embodiment, a single surge 1402 plate is attached externally to the structural envelope, to the bottom of the arm 1404. The surge plate 1402 forms a triangle, with the hypotenuse attached to the bottom of the arm 1404, the long side 1426 running parallel to baseline and the short side 1428 extending vertically downwards from the tendon porch 1422. In one embodiment a keel plate 1430 is attached to support the long side 1426 of the surge plate 1402 along baseline. In one embodiment, the lateral projected area of the surge plate 1402 is equal to 60% of the lateral projected area of the arm 1404 without surge plates. In alternate embodiments, the lateral projected area of the surge plate 1402 is greater than or equal to 25% of the lateral projected area of the arm 1404 without surge plates.

Claims

1. A method of decreasing wave induced accelerations of a tension-leg platform having three or more tendons, the method comprising: increasing a mass of water displaced by the platform when the platform is accelerating horizontally by a system of one or more surge plates connected to a submerged structure of the platform, the surge plates including thin plates and/or panels, oriented to be substantially vertical when the platform is upright, whereby a submerged lateral projected area of the surge plates is greater than or equal to 10% of a lateral projected area of the submerged structure of the platform excluding the surge plates.
2. The method of claim 1, wherein the surge plates are attached to a lower half of the submerged structure of the installed platform when measured at low astronomical tide.
3. The method of claim 1, wherein at least one surge plate is configured to be an open structure, with no enclosed buoyant volumes.
4. The method of claim 1, wherein at least one surge plate is configured to be a closed structure, with enclosed buoyant volumes.
5. The method of claim 1, wherein one or more sides of at least one surge plate is rigidly attached external to a structural envelope of the tension-leg platform.
6. The method of claim 1, wherein one or more sides of at least one surge plate is rigidly attached within a structural envelope of the tension-leg platform.
7. The method of claim 1, wherein one or more sides of a surge plate is rigidly attached to a structural member which extends radially from a central column and supports a tendon at its distal end, the lateral projected area of the surge plate is greater than or equal to 25% of the lateral projected area of the radial structural member excluding the surge plate.
8. A system of one or more surge plates attached to a structure of a tension-leg platform, the surge plates comprising thin plates and/or panels, oriented to be substantially vertical when the platform is upright, whereby a submerged lateral projected area of the one or more surge plates is greater than or equal to 10% of a lateral projected area of a submerged structure of the platform excluding surge plates.
9. The system of claim 8, wherein the surge plates are attached to a lower half of the submerged

structure of the installed platform when measured at low astronomical tide.

10. The system of claim 8, wherein at least one surge plate is configured to be an open structure, with no enclosed buoyant volumes.

11. The system of claim 8, wherein at least one surge plate is configured to be a closed structure, with enclosed buoyant volumes.

12. The system of claim 8, wherein one or more sides of at least one surge plate is rigidly attached external to a structural envelope of the tension-leg platform.

13. The system of claim 8, wherein one or more sides of at least one surge plate is rigidly attached within a structural envelope of the tension-leg platform.

14. A floating, tension-leg wind turbine platform comprising: a central cylindrical column; three or more arms, equally spaced circumferentially around and extending radially from a base of the central column, for each arm, one or more tendon porches located at a distal end of each arm for attaching one or more tendons; for each arm one or more surge plates including thin plates and/or panels oriented to be substantially vertical when the platform is upright and rigidly attached to each arm, a sum total of a lateral projected area of the surge plates of each arm is greater than or equal to 25% of a lateral projected area of the arm excluding the surge plates; a wind turbine tower attached to the top of the central column such that the long axis of the tower is parallel to the long axis of the central column.

15. The floating wind turbine platform of claim 14, wherein the central column is constructed of a single constant diameter cylinder.

16. The floating wind turbine platform of claim 14, wherein the column is constructed of segmented cylinders of different diameters serially connected together by tapered conical sections, a bottom most cylinder having a largest diameter and diameters of remaining cylinders incrementally decreasing with height.

17. The floating wind turbine platform of **15** or claim 16, wherein each arm is arranged with a vertically oriented cylindrical buoyancy can supported at a distance radially from the central column by a vertically oriented planar truss, the top and bottom chords of the truss constructed from large-diameter cylinders, each chord having a first end for connecting to the central column and a second end for connecting to the buoyancy can, the chords supported near their mid-spans by a single cylindrical vertical cylinder; and each one of the surge plates is attached to the central column or respective buoyancy can, and respective truss members, filling at minimum 50% of an open area within each respective arm truss.

18. The floating wind turbine platform of claim 15 or claim 16, wherein each arm forms a polyhedron approximately shaped like a truncated wedge with outer surfaces formed by flat panels, a bottom of the arm aligned with a bottom of the platform and parallel to baseline, a top of the arm sloping down such that a tip of the arm is shorter in height than a root of the arm; and a surge plate is attached to the top of the arm, the lateral projected area of the surge plate is greater than or equal to 25% of the lateral projected area of the arm excluding the surge plate.

19. The floating wind turbine platform of claim 15 or claim 16, wherein each arm forms a polyhedron approximately shaped like a truncated wedge with its outer surfaces formed by flat panels, the top of the arm aligned parallel to and at a distance above the baseline of the platform, the bottom sloping up from baseline such that a tip of the arm is shorter in height than the root of the arm; and a surge plate is attached to the bottom of the arm, the lateral projected area of the surge plate is greater than or equal to 25% of the lateral projected area of the arm excluding the surge plate.

20. The floating wind turbine platform of claim 18, wherein sides of the arms are angled inwards such that the tip of each arm is narrower than the root.

21. The floating wind turbine platform of claim 19, wherein sides of the arms are angled inwards such that the tip of each arm is narrower than the root.

22. The floating wind turbine platform of claim 18, wherein sides of the arms are parallel such that

the tip of each arm is the same width as the root.

23. The floating wind turbine platform of claim 19, wherein sides of the arms are parallel such that the tip each arm is the same width as the root.
